AI-Powered Education Student Success Assistant

End-to-End System Architecture: Data Ingestion to Multi-Agent Decision Support & Continuous Feedback

1. Dataset
OULAD Open University

- 7 Relational CSV Tables
- 32,593 Student Enrolments
- Assessments & VLE Clicks
- Course Module Schedules

2. Preprocessing
Temporal Scoping & Feature Store

- Multi-checkpoint Cutoffs (w6/9/15/22)
- In-Scope Filtering (Drop Early Leaves)
- Weekly Grid Aggregations
- Sensitive Feature Auditing

3. Chunking
Qualitative Behavioural Tokenization

- Discretized Cohort Bands
- Tokens: score_low, missed_most
- Engagement Momentum Tokens
- Student Case Summaries

4. Embeddings
Supervised & Dense Representation

- Model Attribution-Weighted Vectors
- TreeSHAP-Scaled Feature Vectors
- MiniLM Dense Narratives
- Lexical Token Bags

5. Vector Database
Case Index & Metadata Storage

- Supervised kNN Index
- BM25 Inverted Index
- 25k+ Historical Student Cases
- SQLite Metadata & Feedback DB

Historical Precedents

6. Retriever
Hybrid Search Engine

- Keyword (BM25) + kNN Vectors
- Reciprocal Rank Fusion (RRF)
- Metadata Filters (Course, Region, etc.)
- 'What Happened Next' Stories

7. LLM / RAG Pipeline
Context Grounding & Guardrails

- Evidence Assembly & Context Pack
- Code Guardrails (Numeric Grounding)
- Precedent Synthesis Engine
- Stigma & Punitive Term Screening

8. Multi-Agent System
Collaborative Student Support

- Performance Analysis Agent
- Risk Classification Agent
- Action Router & Intervention Agent
- Critic Agent (1x Revision Loop)

9. API Layer
FastAPI Microservice Endpoints

- REST Endpoints (/analyze, /predict)
- API-Key Auth & Rate Limiting
- Background Jobs with Step Streaming
- Strict Pydantic Contracts

10. User Interface
Educator Decision Consoles

- Interactive Streamlit App (frontend.py)
- React 18 Executive Dashboard
- Course Clock & Interactive Timeline
- Live Copilot Q&A & Outreach Generator

Continuous Feedback Loop: Advisor Ratings, Plan Acceptance, Real Outcomes & Model Recalibration